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Imaging and Differentiation of Retinal Ganglion Cells in Ex Vivo Experimental Optic Nerve Degeneration by Differential Interference Contrast Microscopy

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ABSTRACT

Purpose: Apoptotic loss of retinal ganglion cells (RGCs) is involved in various optic neuropathies, and its extent is closely related to visual impairment. Direct imaging and counting of RGCs is beneficial to the evaluation of RGC loss, but these processes are challenging with the conventional techniques, due to the transparency and hypo-reflectivity of RGCs as light-transmitting structures of the retina. Differential interference contrast (DIC) microscopy, which can provide real-time images of transparent specimens, is utilized to image neuronal cells including RGCs in the ganglion cell layer (GCL).

Methods: Herein, we show that the neuronal cells within each GCL in an explanted rat retina, including the inner nuclear layer and the outer nuclear layer, can be imaged selectively by transmission-type DIC microscopy. RGCs were also differentiated from non-RGCs by the objective method.

Results: RGCs were differentiated from non-RGCs in the GCL by their morphological features on DIC images with the aid of retrograde fluorescence labeling. Loss of RGCs was detected in optic-nerve-transection and retinal-ischemia-reperfusion models by DIC imaging. The images obtained from the reflection-type DIC microscopy were comparable to those from the transmission-type DIC microscopy.

Conclusions: This method enables direct optical visualization of RGCs in experimental optic-nerve degeneration, thus providing the opportunity for more accurate evaluation of optic neuropathies as well as more effective investigation of diseases.

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Introduction

The specially functionalized cell layers in light-sensing retinal tissue include the ganglion cell layer (GCL) and the photoreceptor layer. Light passes through the transparent retinal layers to reach the photoreceptors. The photoreceptors convert light to bioelectrical signals and transfer visual signals back to the retinal ganglion cells (RGCs) in the GCL. The axons of the RGCs form the optic nerve, which relays visual signals to the brain. RGCs are the only retinal cells involved in the transmission of visual signals to the brain. RGC degeneration is observed in many optic neuropathies including glaucoma as well as hereditary, ischemic and traumatic optic neuropathies.¹ RGC apoptosis is involved in these diseases^{2–4}, the extent and location of RGC loss being closely associated with visual-functional impairment.^{5,6} Certainly, direct imaging of RGCs and their density would provide much more accurate evaluation of optic neuropathies and enable the investigation of diseases as well. However, since RGCs are transparent to and hypo-reflective of light, conventional imaging techniques including optical microscopy are unsuitable for their visualization.

Differential interference contrast (DIC) microscopy is designed to optically image unstained, transparent specimens by enhancing the contrast that reflects the Wollaston prism's induced interference of the two orthogonal beams with the offset occurring gradient of the optical path length. This methodology gives DIC microscopy a key advantage over other imaging modalities, and indeed it is widely utilized to image a variety of transparent specimens.^{7–10} In our previous paper, we demonstrated DIC microscopy imaging of progressive morphological changes to retinal photoreceptor cells in a rodent retinitis pigmentosa model.¹¹

In the present study, we employed, initially, transmission-type DIC microscopy in order to image RGCs in a rat retina. By modulating the wavelength and bandwidth of the light source with a tunable band-pass filter, our DIC microscopy could image RGCs and objectively differentiate them from other neuronal cells including displaced amacrine cells in the GCL. We were able to detect sequential RGC loss as disease progression in rat optic-nerve-degeneration models. Subsequently, we also set up reflection-type DIC microscopy and compared the images obtained thereby with the

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transmission-type DIC microscopy images. This could be a useful technique for evaluating experimental optic-nerve degeneration.

Materials and methods

Differential interference contrast (DIC) microscopy system

DIC microscopy is an imaging technique that utilizes beam interference for unstained and transparent specimens. (Figure 1) The front linear polarizer converts the unpolarized beam, the bandgap and wavelength of which are modified, into a 45°-polarized beam. The controlled beam meets the Wollaston prism, which separates the light wave into two directions with orthogonal polarization. The condenser refracts the orthogonal beams to illuminate the specimen, and pass through it with a small offset. When the orthogonal offset beams contact the transparent specimen, their velocities are different. Each beam experiences a different thickness and refractive index, which induce different optical path lengths. Thus, the beams contain gradient information, in the form of a phase shift, between the offset. For a transmission-type DIC microscopy, the second Wollaston prism and the second linear polarizer are located behind the specimen and attune the beams' paths and polarizations. The resultant combined beam shows interference due to the phase difference. The bandpass filter is utilized to tune the wavelength (790 nm) and

bandwidth (1.5 nm). A second condenser lens is aligned for Kohler illumination. Additionally, fluorescence microscopy is combined with DIC microscopy to confirm the RGCs in the identical area in the layer. To construct a reflection-type DIC, a beam splitter (50:50) was placed between the objective lens and the Wollaston prism. (Figure 1e)

Animals and retina preparation

The use of animals for this study was approved by the Animal Research Committee of Seoul National University Hospital and complied with the Association for Research in Vision and Ophthalmology's Statement on the Use of Animals in Ophthalmic and Vision Research. Male Sprague-Dawley rats weighing 250 to 300 g were housed with standard chow and water provided *ad libitum*. The procedure used to generate the optic-nerve transection (ONT) model has been described previously.¹² Briefly, the animals were anesthetized by intramuscular injection of xylazine (Rompun, Bayer, Germany; 1 mg/kg) and Zoletil (Virbac Animal Health Ltd, France; 10 mg/kg). The lateral conjunctiva was incised with scissors, and the optic nerve was exposed by blunt dissection. A 2-mm longitudinal incision starting 3 mm behind the eye was made in the optic nerve sheath with a needle knife. A cross-section of the optic nerve was made with the needle knife through the opening of the

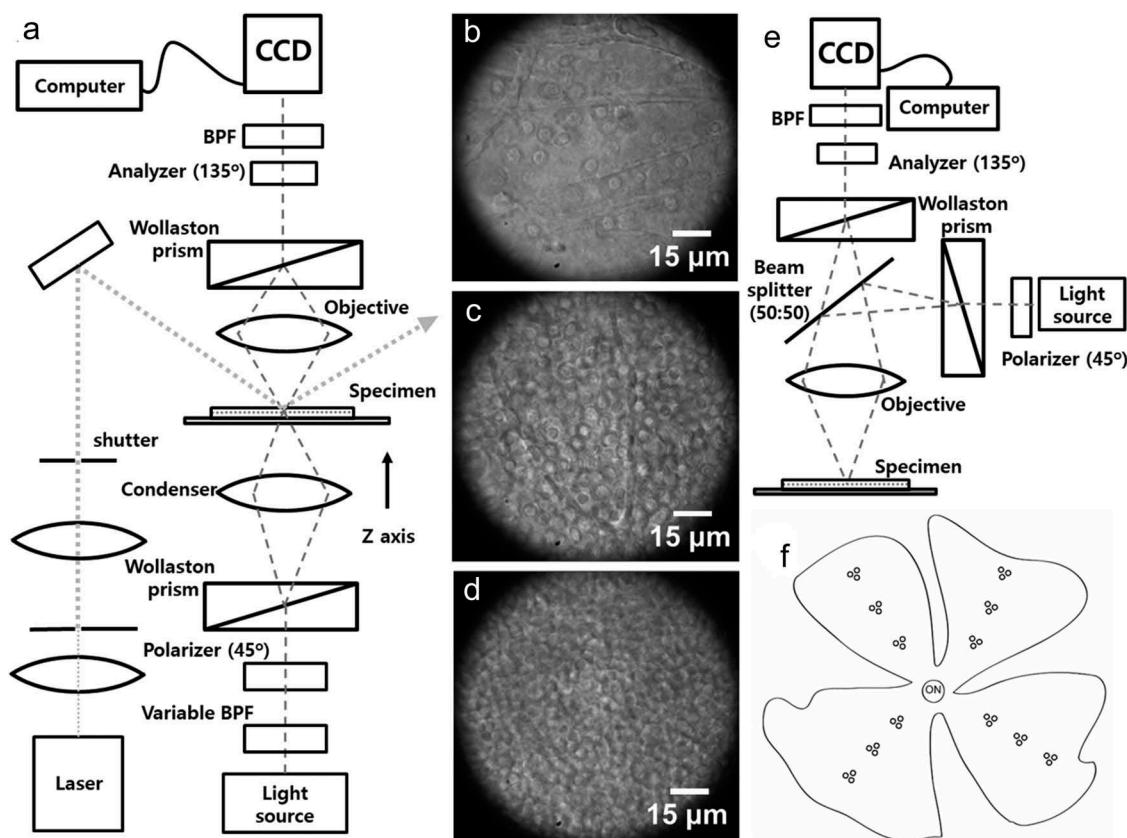


Figure 1. Experimental set-up and DIC images of retina according to depth of focus. (a) Transmission-type DIC and fluorescence microscopy verifying RGCs in GCL. The dark dashed lines are the DIC beam path, and the bright short-dashed line represents the fluorescence microscopy. The retina was imaged using transmission-type DIC, and (b) the GCL, (c) the inner nuclear layer and (d) the outer nuclear layer were imaged according to the depth of focus. (e) Reflection-type DIC microscopy (f) In a retinal flat mount, images were captured in three adjacent fields, near, mid, and far from the center of the optic nerve, respectively, along the centerline of each retinal quadrant.

optic nerve sheath, with care taken to avoid damage to the adjacent blood supply. The conjunctival incision was sutured, and antibiotic ophthalmic ointment was applied topically. ONT was performed on the left eye of each rat. After completion of the ONT procedure, dextran tetramethylrhodamine (DTMR, 3000 MW; Molecular Probes, Eugene, OR) was immediately applied to the proximal cut surface of the optic nerve for retrograde labeling of RGCs. Eyes were enucleated at post-ONT 1 day, 1 week and 2 weeks, and the retinas were dissected from the eye cups and prepared as flat mounts, with 4 radially oriented cuts in each retina. The retinas were then whole-mounted on glass slides. A post-ONT 1-day retina was regarded as the control. Application of the retinal-ischemia-reperfusion model entails the following procedure. A steel 30-gauge infusion needle connected to a pressure device, elevated to 160 mm Hg, was inserted into the middle of the anterior chamber of the eye. Successful retinal ischemia was indicated by the collapse of the central retinal artery and whitening of the iris with IOP elevation. The IOP was elevated at a sustained 160 mm Hg for 75 min; reperfusion was allowed for a period of 1 and 2 weeks, respectively, for the two experimental groups. In the sham-operated control group, the infusion needle was inserted into the anterior chamber of the eye

without causing elevated IOP. Retrograde fluorescent labeling of RGCs was not performed. The eyes were enucleated, and their retinas were flat-mounted.

Image analysis and cell counting

Using the transmission-type DIC microscopy, we observed the retinal cells in flat-mounted retinas. In order to identify the GCL, fluorescence images were compared with DIC images with the x-y axis fixed. Then, we imaged GCL, inner nuclear layer and outer nuclear layer. For quantification of neuronal cells in GCL, images were captured in three adjacent fields of 1 mm (near), 2 mm (mid), and 3 mm (far) respectively from the center of the optic nerve along the centerline of each retinal quadrant. (Figure 1f) The DTMR-labeled RGCs were imaged under DIC microscopy and fluorescence microscopy alternatively for the same area. Thus, a total of 72 images (36 DIC images and 36 fluorescent images) were obtained for each retinal sample. For evaluation of total neuronal cells in GCL, the number of any cells in each 0.01 mm² region of DIC images (Figure 2, equal to the area of field of view) was counted by 2 masked investigators and averaged. To evaluate the morphological features of RGCs in DIC microscopy, the images of DTMR-labeled RGCs were compared between fluorescence microscopy and DIC microscopy.

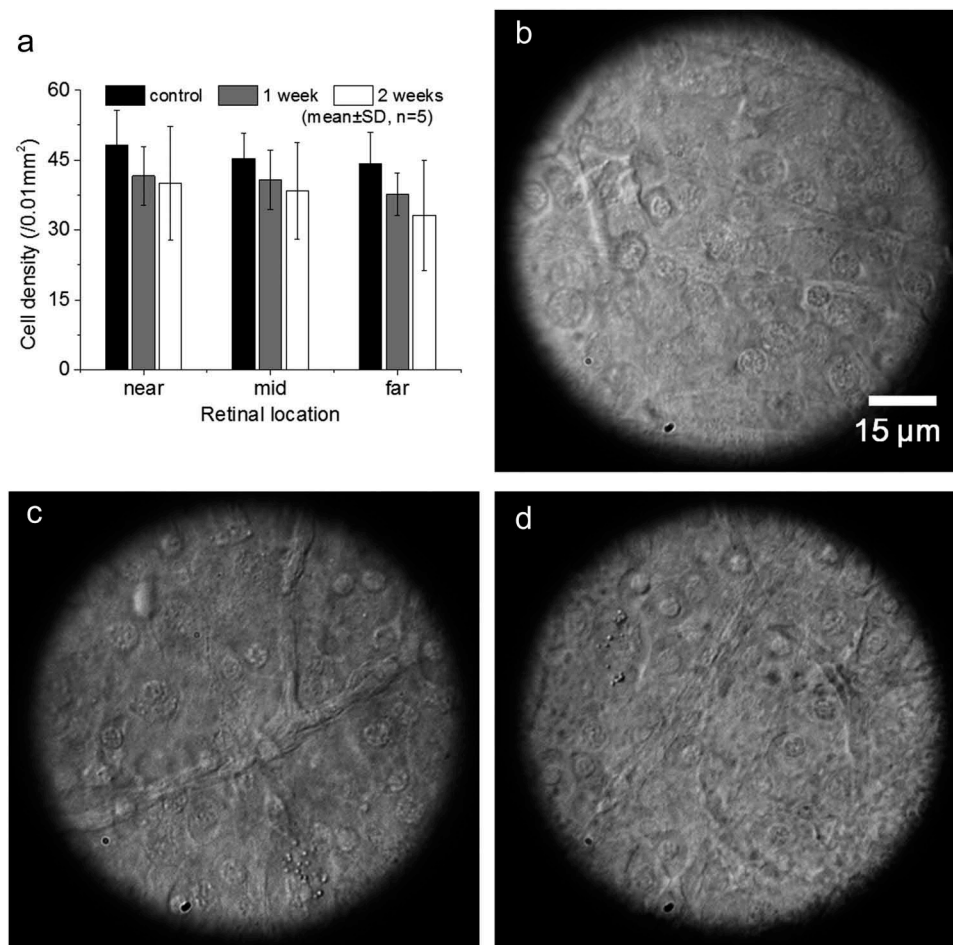


Figure 2. Total cell number of GCL in ONT model. (a) Total number of cells in GCL counted in ONT model according to location and disease progression. (b) GCL imaged using DIC for control, (c) at post-ONT 1 week, and (d) at post-ONT 2 weeks.

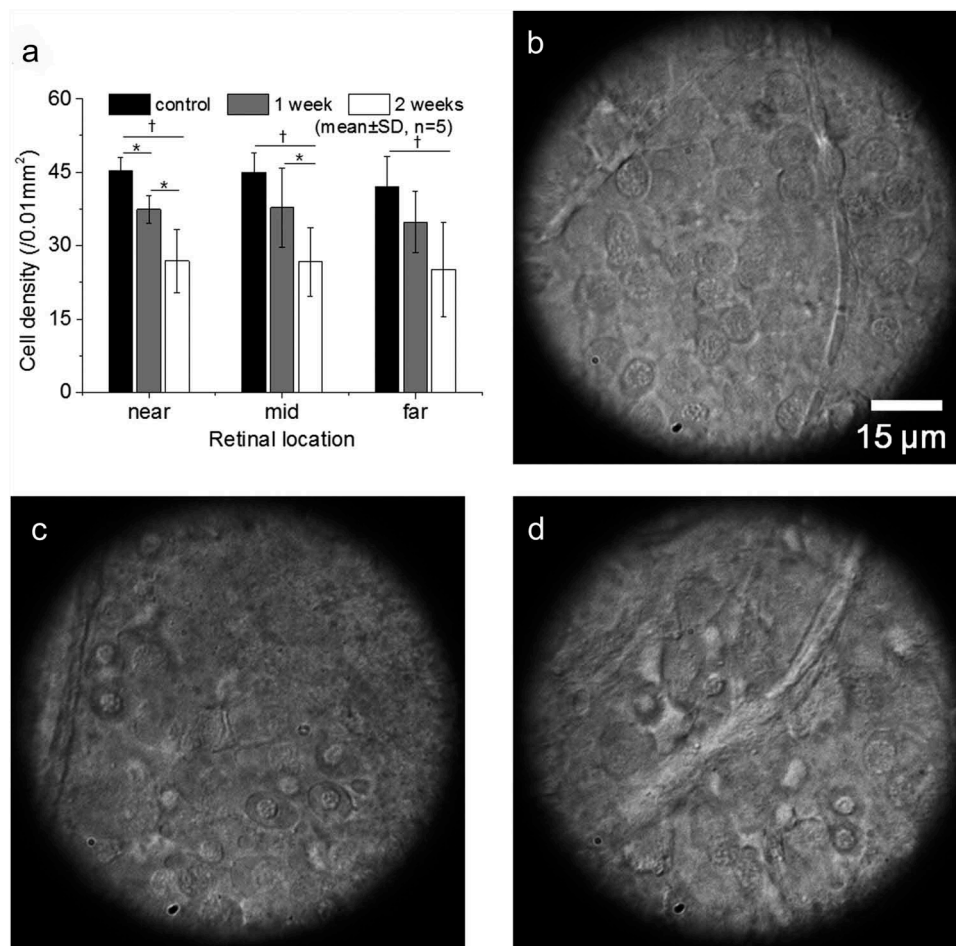


Figure 3. Total cell number of GCL in ischemia-reperfusion model. The total number of cells was counted in the ischemia-reperfusion model according to location and disease progression. (a) The number of samples and points are identical to those of the ONT model. The GCL was imaged using DIC for (b) control, (c) 1 week and (d) 2 weeks, respectively (*: $p < 0.05$, †: $p < 0.01$, One way-ANOVA, post-hoc LSD test).

To differentiate RGCs from non-RGCs objectively, cells in the images obtained by DIC were evaluated. Since the intensity value at the boundaries of the nucleus and cytoplasm changes drastically, the diameter of both parts for each cell was noticeable. To measure the cross-sectional distance of the parts in each cell, the intensity values from DIC were extracted. As shown in Figure 4, the diameters of the nucleus (r_n) and cytoplasm (r_c) for both cells were measured from the extracted intensity values. The dark and light gray bars represent the diameters of the cytoplasm and nucleus, respectively. After measuring the cross-sectional diameter of both parts, the ratio (r_c/r_n) was calculated by dividing the diameter of the cytoplasm by that of the nucleus. Ninety-three (93) RGCs and 94 non-RGCs, a total of 187 cells, were extracted from arbitrary images, the relative size ratio values were obtained, and the threshold values were set.

To quantify RGCs in DIC images, investigators have used morphological criteria and, if needed, the objective method. The morphologic criteria of RGCs in DIC images are as follows: a less-clear nucleus/cytoplasm boundary, a relatively large size, and a smaller cytoplasm-to-nucleus ratio. Because it was time-consuming to apply the objective method to all of the cells on the images, investigators identified RGCs by the morphological criteria for most cells and applied the objective method for ambiguous cells. For the validation of the

application of semi-objective differentiation of RGCs from non-RGCs in the GCL, a masked investigator counted the RGCs in 30 randomly selected DIC images, with recourse to the objective method if needed, and without reference to the relevant fluorescence images, and the results were compared with those for the fluorescence images, which image type is regarded as the gold standard for RGC counting. On that basis, the sensitivity and specificity of the morphological differentiation of RGCs from non-RGCs were calculated.

For evaluation of the number of RGCs and non-RGCs in experimental optic-nerve-degeneration models, five retinal samples were imaged for each disease stage. The numbers of cells in 12 photographs for each location (near, mid, far) were obtained. Since the wavelengths utilized in DIC and fluorescence microscopy are different, the focusing point along the z axis was offset. After matching the DIC images to the fluorescence images, the RGCs in the GCL were manually counted. In the ONT model, the counting process was conducted by reference to the fluorescence images. In the retinal-ischemia-reperfusion model, the counting process was conducted only with the DIC images. The RGCs were semi-objectively counted by 2 investigators in a masked fashion and averaged. The number of non-RGCs was obtained by subtraction of the number of RGCs from the total number

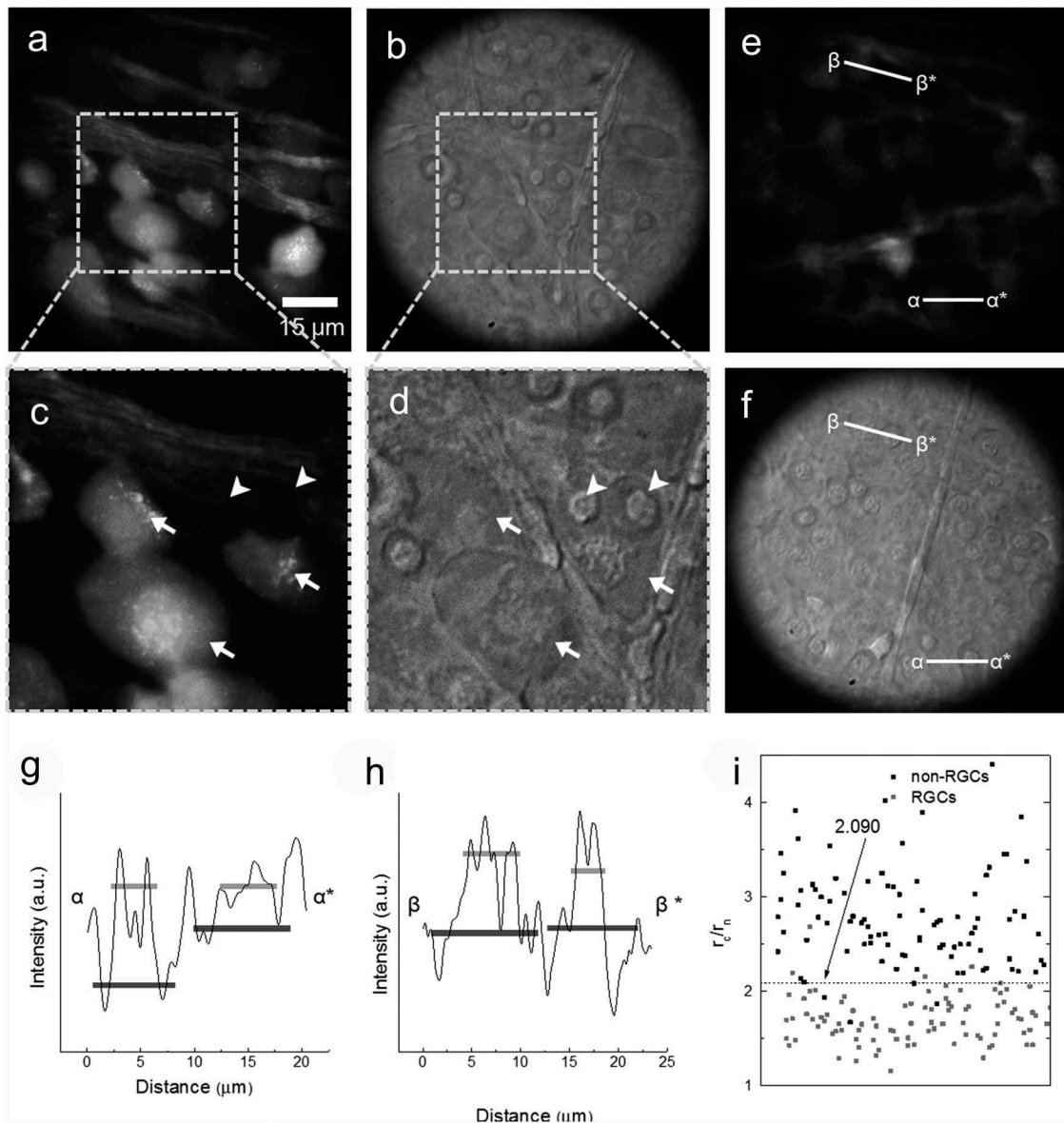


Figure 4. Differentiation and objective method for differentiation of RGCs from non-RGCs in GCL. The RGCs could be confirmed on (a) retrogradely labeled fluorescence imaging and (b) DIC imaging of GCL in identical position. Comparison of the RGCs with non-RGCs could be made by (c) fluorescence imaging and (d) DIC imaging of the enlarged box sections (indicated in (a) and (b)). The arrow heads indicate amacrine cells, and the arrows indicate RGCs. To obtain an objective index using the ratio of diameters, cells were randomly selected from (e) fluorescent images and (f) DIC images. The diameters of the cytoplasm (r_c) and nucleus (r_n) for both cells were measured from the extracted intensity values in (g) and (h). The dark- and light-gray bars represent the diameters of the cytoplasm and nucleus, respectively. (i) Distribution of RGCs (gray) and non-RGCs (black) by diameter ratio with the cut-off value of 2.090.

of cells in the same image. The counted cells were statistically analyzed by one-way ANOVA and post-hoc LSD tests (SPSS version 22.0, Chicago, IL).

Results

Imaging of total neuronal cells in retinal ganglion cell layer by DIC microscopy

We were able to optically image neuronal cells selectively in the GCL, and inner and outer nuclear layers by DIC microscopy (Figure 1, SI Movie 1). In an ONT model, the overall number of total neuronal cells in the GCL was decreased with disease progression, but the extent of decrease did not reach statistical significance (Figure 2). For the near, mid and far points from the

center of optic nerve in the control group, the total numbers of cells in the retina were 48.2 ± 7.4 , 45.4 ± 5.5 and 44.2 ± 6.8 (mean $\pm$ SD, $n = 5$) in each 0.01 mm^2 region, and the total numbers of neuronal cells were 40.1 ± 12.1 , 38.4 ± 10.4 and 33.1 ± 11.8 (mean $\pm$ SD, $n = 5$) at post-ONT 2 weeks, respectively. In the retinal-ischemia-reperfusion model, for the near, mid and far points, the total numbers of cells in the GCL significantly decreased from 45.3 ± 2.8 , 45.0 ± 3.9 and 42.1 ± 6.2 (mean $\pm$ SD, $n = 5$) to 26.9 ± 6.5 , 26.7 ± 7.0 and 25.2 ± 9.6 (mean $\pm$ SD, $n = 5$) after 2 weeks, respectively (Figure 3).

Differentiation of RGCs from non-RGCs

Because RGCs as well as displaced amacrine cells and microglial cells are located in the GCL, differentiation of RGCs

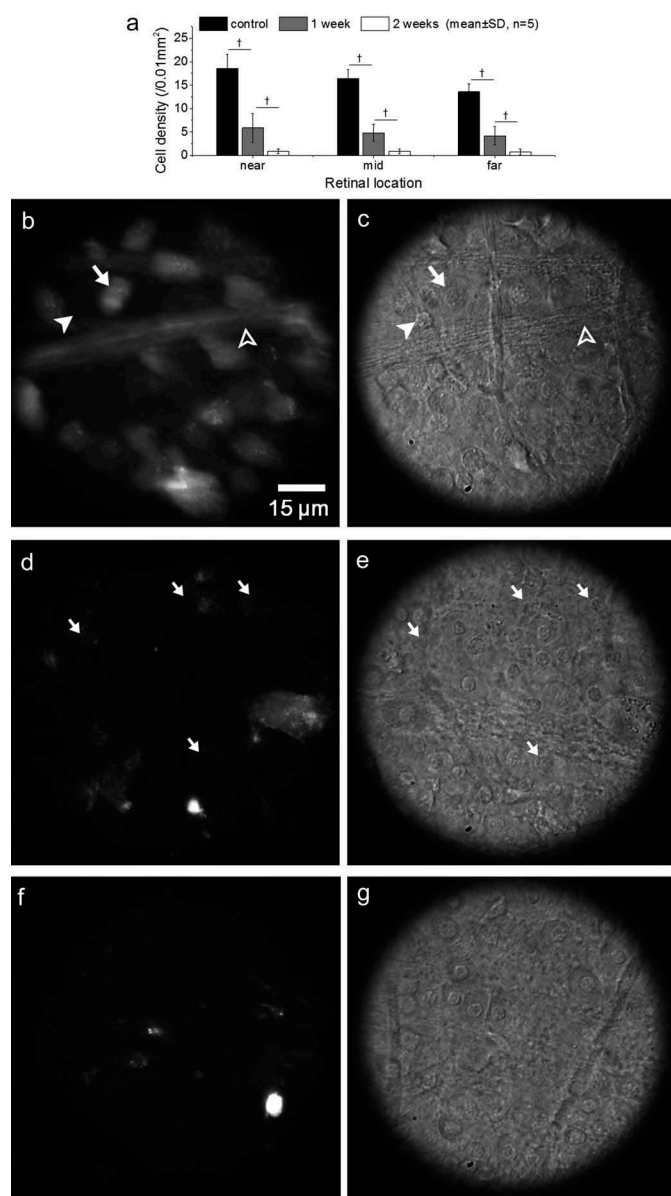


Figure 5. Number of RGCs in ONT model. (a) The total number of RGCs in the GCL was counted for the control and ONT model according to location and disease progression. The images show fluorescence and DIC images of the GCL layer (b, c) for the control, (d, e) post-ONT 1 week and (f, g) post-ONT 2 weeks. As shown in (b) and (c), the RGC (solid arrow), amacrine cells (solid arrow head) and RNFL (framed arrow head) were well imaged. As seen in (f) and (g), after 2 weeks, most of the RGCs had disappeared, and only the morphologically apparent amacrine cells were visible. (†: $p < 0.01$, One way-ANOVA, post-hoc LSD test).

from non-RGCs, which are mainly displaced amacrine cells, is mandatory in any RGC-density measurement. DIC microscopy provides a sufficiently high resolution for imaging of the cell border and cytoplasm as well as the nucleus, thereby enabling such differentiation (Figure 4). Comparison of the DIC images with fluorescence images indicated that the RGCs had a less-clear nucleus/cytoplasm boundary, a relatively large size, and a smaller cytoplasm-to-nucleus ratio than the non-RGCs in the GCL. There were no differences in RGC morphology according to the near, mid and far regions. In order to objectively differentiate RGCs from non-RGCs, we applied the r_c/r_n criterion, and found that the RGCs were excellently

differentiated from the non-RGCs (AUROC: 0.981). The sensitivity and specificity were both 93.75% at the cut-off value of 2.090 (Figure 4).

To determine the diagnostic accuracy of semi-objective differentiation of RGCs from non-RGCs, we compared the RGC count results obtained without the help of fluorescence images by applying the morphological criteria of DIC images and (if needed) the objective method as well as the counts obtained with the aid of fluorescence images (gold standard). According to the two sets of results, the semi-objective interpretation of DIC microscopy showed 82.4 and 91.8% RGC-identification sensitivity and specificity, respectively.

Change of RGC density in optic-nerve-degeneration animals

As shown in Figure 5, the number of RGCs significantly decreased over time in the ONT model: from 18.6 ± 3.0 , 16.4 ± 1.9 and 13.6 ± 1.7 (mean $\pm$ SD, $n = 5$) at each location (near, mid and far) in the control eyes to 0.9 ± 0.5 , 0.9 ± 0.5 and 0.7 ± 0.6 (mean $\pm$ SD, $n = 5$) at post-ONT 2 weeks, respectively. The ischemia-reperfusion model showed a similar tendency: the number of RGCs decreased from 17.0 ± 1.5 , 14.7 ± 0.7 and 13.8 ± 1.5 (mean $\pm$ SD, $n = 5$) to 0.4 ± 0.2 , 0.5 ± 0.4 and 0.4 ± 0.3 (mean $\pm$ SD, $n = 5$) at 2 weeks, respectively (Figure 6). There was no change in the shape or size of surviving RGCs at 2 weeks in either model. Additionally, we counted the non-RGCs in both models: there was no statistically significant change with optic nerve degeneration in either model (Figure 7).

Comparison of images of transmission type and reflection-type DIC microscopy

As shown in Figure 8, reflection-type DIC microscopy imaged neuronal cells. The images obtained from the reflection-type DIC microscopy were comparable to the images from the transmission-type DIC microscopy.

Discussion

Using DIC microscopy, we imaged neuronal cells selectively in the GCL. With the aid of fluorescence labeling, we optically identified RGCs on DIC images and detected their morphological features. We also demonstrated the objective and semi-objective methods for differentiation of RGCs from non-RGCs. Then, we quantified the loss of RGCs in ONT and retinal-ischemia-reperfusion models by differentiating them from non-RGCs. Direct imaging of RGCs and measurement of RGC density by noninvasive optical imaging would be helpful for investigation of experimental RGC-degenerative diseases.

Traditionally, for direct imaging and identification, transparent RGCs need to be manipulated for fluorescence. Retrograde fluorescent labeling via the superior colliculus or transected optic nerve stump can label more than 96% and up to 100% of RGCs.^{13–15} However, it requires invasive surgery, and moreover, other cells in the RGC layer such as microglial cells are frequently labeled due to the phagocytosis of

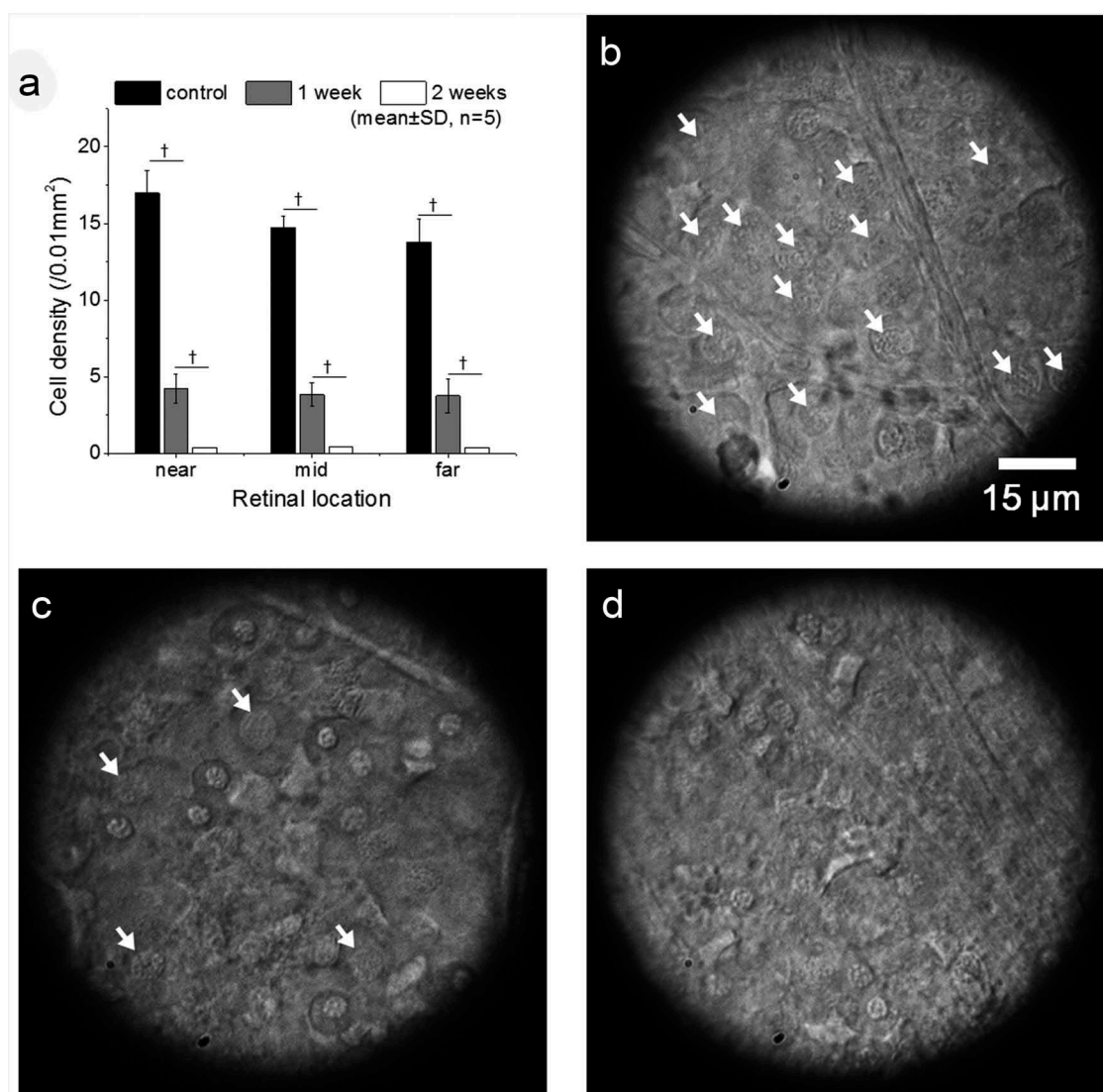


Figure 6. Number of RGCs in ischemia-reperfusion model. The total number of RGCs in the GCL was counted for the control and ischemia-reperfusion model according to location and disease progression. Shown are DIC images of the GCL layer for (b) the control, (c) 1 week, and (d) 2 weeks after the ischemia-reperfusion model was applied. As shown in b, the RGCs (solid arrows) and amacrine cells were well imaged when the disease model was not applied. (c) shows the GCL 1 week after application of the ischemia-reperfusion model. As seen in (d), after 2 weeks, most of the RGCs had disappeared. (†: $p < 0.01$, One way-ANOVA, post-hoc LSD test).

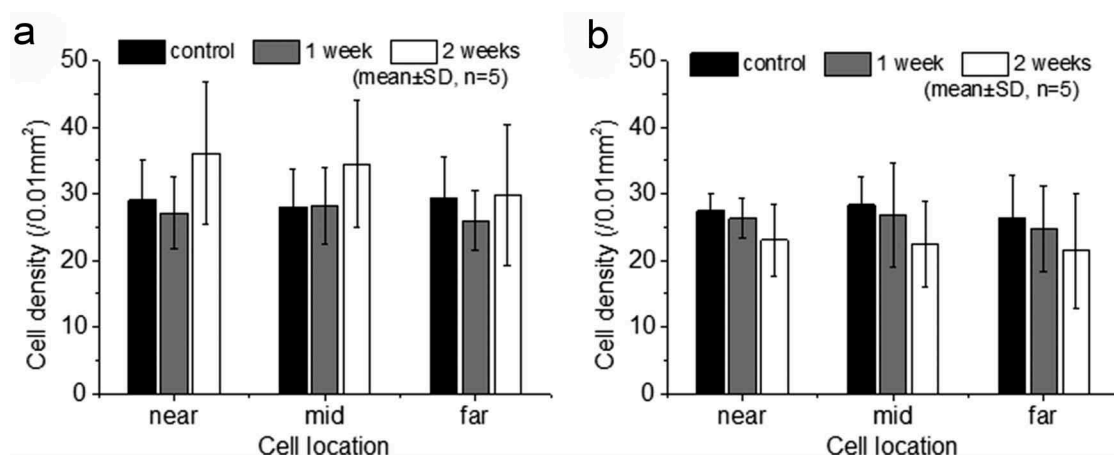


Figure 7. Numbers of non-RGCs in GCL. The numbers of neuronal cells in the GCL (except RGCs) were counted in (a) the ONT model and (b) the ischemia-reperfusion model. In both models, the number of non-RGCs did not change.

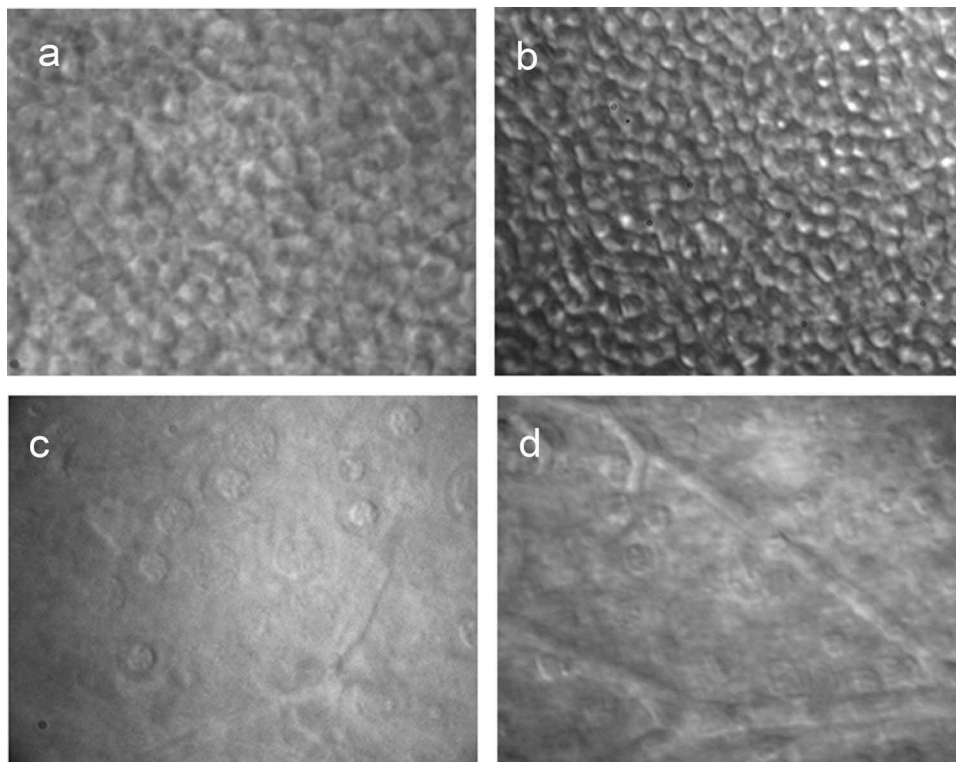


Figure 8. Image comparison according to DIC type. Outer nuclear layer (a, b) and GCL (c, d) images of normal rat retina by DIC microscopy. The images were obtained by (a, c) transmission-type DIC microscopy and (b, d) reflection-type DIC microscopy.

damaged RGCs. Also, the cytoplasm and nucleus cannot be differentiated by fluorescent microscopy. Electroporation of labelling agents is another means of labelling RGCs,¹⁶ but it is not specific to RGCs and can possibly damage ocular tissues. Recently, intravitreal injection of fluorescent agents for labelling of RGCs was reported.¹⁵ However, it requires invasive injection and cannot label all RGCs in the retina. Another limitation of fluorescence labeling is that it is difficult to apply to human-subject studies.

To overcome the limitations of the fluorescent labeling and to extend its application to human eyes, several techniques for optical imaging of RGCs have been introduced. Optical coherence tomography (OCT) can measure the thickness of the RGC axon bundle (the RNFL) but cannot image RGCs directly, due to their low reflectivity and the insufficient transverse resolution of OCT.¹⁷ Recent adaptive-optics OCT can image RGCs, but it is time-consuming and requires averaging of hundreds of en-face images.¹⁸ In addition, it cannot differentiate RGCs from non-RGCs. Adaptive optics-scanning laser ophthalmoscopy (AO-SLO) cannot image RGCs directly, although recently a multi-offset modification of AO-SLO could detect neuronal cells in GCL *in vivo*.¹⁸ Even so, the resolution was not high enough to differentiate the RGCs from displaced amacrine cells. In the present study, we imaged RGCs and their subcellular structures without fluorescence in real time and differentiated RGCs from non-RGCs. Although our results thus far are limited to retinal flat mounts, our method has the potential for extension to human eyes.

According to our ONT model, the RGC density was decreased from 30 to 32% at post-ONT 1 week and to 5% at 2 weeks. This result is comparable to the data of previous studies: 50 to 65% RGC loss by post-ONT 1 week and over 90% by 2 weeks.^{19,20} The total cell number was not statistically changed, and the non-RGCs were slightly increased after 2 weeks though not to statistically significant levels. This increase in non-RGCs might be attributable to microglial proliferation (especially as displaced amacrine cells do not increase after ONT.²¹) Microglial cells are retina-resident phagocytic cells that begin proliferation at post-ONT 1 week and peak during post-ONT weeks 2 to 3.^{22,23} In our study, however, we did not differentiate microglial cells among non-RGCs. The loss of RGCs in our ischemia-reperfusion model was also similar to previous results.²⁴ For a previous ischemia-reperfusion model, amacrine cells were reported to be decreased significantly.²⁵ In our results, non-RGCs decreased, but not to a statistically significant extent. We postulate again that microglia proliferation might explain this. The small sample size in this study also might have affected the result. In our study, RGCs comprised up to 40% of the total cells in the GCL, which ratio is comparable to that obtained in a previous study, in which, among the neuronal cells of the GCL, 40% were RGCs, 50% were displaced amacrine cells, and 10% were glial cells.²¹ Histologically, displaced amacrine cells are of relatively small size, and have little cytoplasm. RGCs, contrastingly, are of a relatively large size and have larger somata compared with amacrine cells.²⁶ Our RGC-morphological criteria coincide with these findings. However, we were not able to

differentiate microglia or astrocytes from displaced amacrine cells. Further study is required to identify those cells.

To objectively differentiate RGCs from non-RGCs, we compared the relative size ratio values (r_c/r_n) obtained by DIC. According to the threshold value of 2.090, highly accurate classification was performed. However, because this method is time-consuming, we were not able to apply it to all of the images. To increase the accuracy while reducing the time, we applied the subjective method to the first judgment and applied the objective method only to the ambiguous cells. This semi-objective method showed relatively good sensitivity and specificity. Subjective differentiation by only morphological criteria showed 66.7% sensitivity and 80.8% specificity (data not shown). Further study is warranted for automatic application of the objective method to all images.

DIC microscopy produces a contrast in the form of the optical path difference between two beams that are split in the Wollaston prism. The difference of the refractive indices generates the optical path difference, and the interference of the two lights is converted to visible changes. A small phase difference is sufficient to generate contrast; thereby, the cell margin as well as the subcellular organelles can be discriminated. In this study, our analysis was performed with the transmission-type DIC microscopy system. For *in vivo* retinal imaging, however, reflection-type DIC microscopy is mandatory. In addition, to overcome optical aberration from the anterior segment of the eyeball, the adaptive optics technique also is essential. In this study, we also set up and utilized reflection-type DIC microscopy, and the obtained images were comparable. Further study to combine the adaptive optics system and reflection-type DIC microscopy is under investigation for *in vivo* imaging.

This study has some limitations. First of all, we used retrograde labeling of RGCs as a gold standard for identifying RGCs. However, in advanced stages of optic nerve degeneration, microglial cells can also be stained after phagocytizing RGC debris. However, there is no perfect way to visualize rat RGCs with fluorescence²⁶. Second, we quantified RGCs and non-RGCs by transmission-type DIC microscopy, because reflection-type DIC microscopy was only recently established. Considering the image quality of reflection-type DIC microscopy, we believe that the RGC-detection utilities are comparable. Third, RGCs by DIC microscopy had less clear cellular boundaries than other cells in the retina, which sometimes made for difficulty in identifying RGCs. However, we could nonetheless detect RGCs, and in fact, this feature enabled differentiation of RGCs from other cells in the RGC layer.

This study demonstrated the possibility of direct optical imaging and differentiation of neuronal cells in the GCL as well as the observation of the course of neuronal cell loss in optic-nerve-degeneration models. Because RGC loss is directly related to visual-functional defect, this method can allow for investigation of mechanisms of optic nerve diseases. Further studies on the optical imaging of RGCs *in vivo* are required.

Disclosure Statement

There is no conflict of interest. The authors declare no competing financial interests.

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